ETH zürich

D INFK

Probabilistic Artificial Intelligence

Introduction

Prof. Andreas Krause
Learning and Adaptive Systems (las.ethz.ch)
Institute for Machine Learning

What is AI?



“The science and engineering of making intelligent machines”

(John McCarthy, ‘56)



Alan Turing (‘50): Computing Machinery and Intelligence

https://en.wikipedia.org/wiki/John_McCarthy_(computer_scientist)#/media/File:John_McCarthy_Stanford.jpg

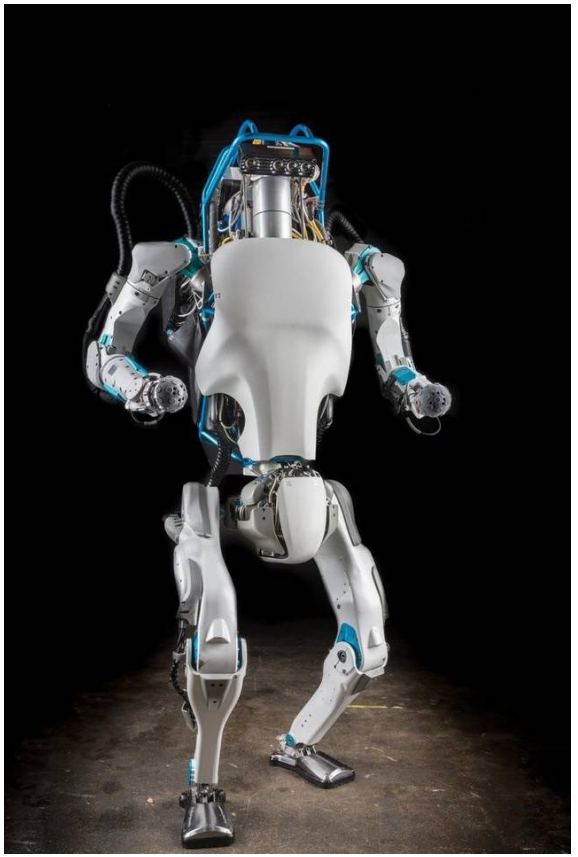
AI as an Engineering Discipline

- Build systems that solve tasks commonly associated with requiring human-level intelligence
- Amenable to engineering principles, empirical evaluation
- Involves / builds on / integrates
  - optimization, algorithms, control theory, formal methods, probability & statistics, game theory, machine learning …
- Much progress on more broadly applicable AI systems
  - While we focus on foundations, we will also cover some key material relevant to generative AI and agents

Humanoid Robotics



Honda ASIMO



Boston Dynamics Atlas

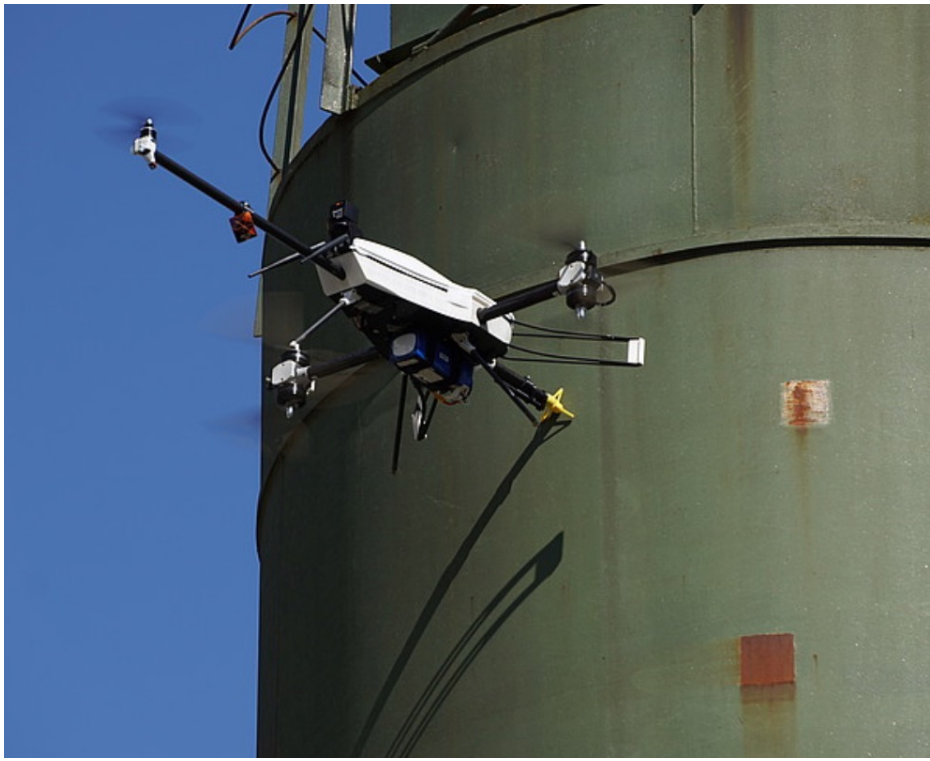
http://www.kansascity.com/news/business/technology/917xpi/picture62197987/ALTERNATES/FREE_640/atlas%20from%20boston%20dynamics

Robotics



ANYmal

[rsl.ethz.ch]

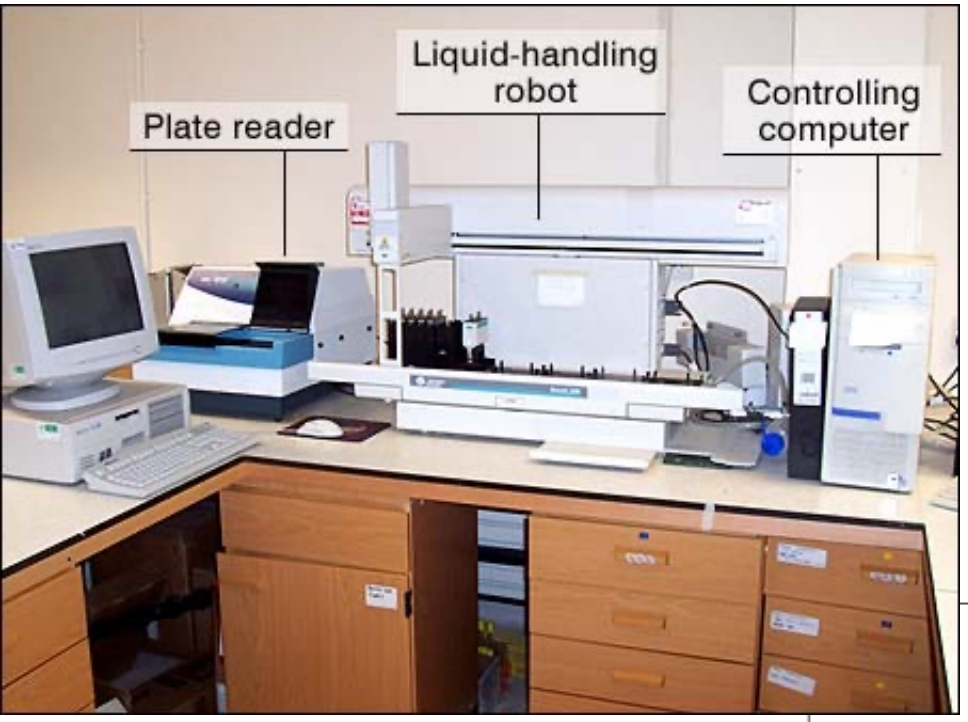


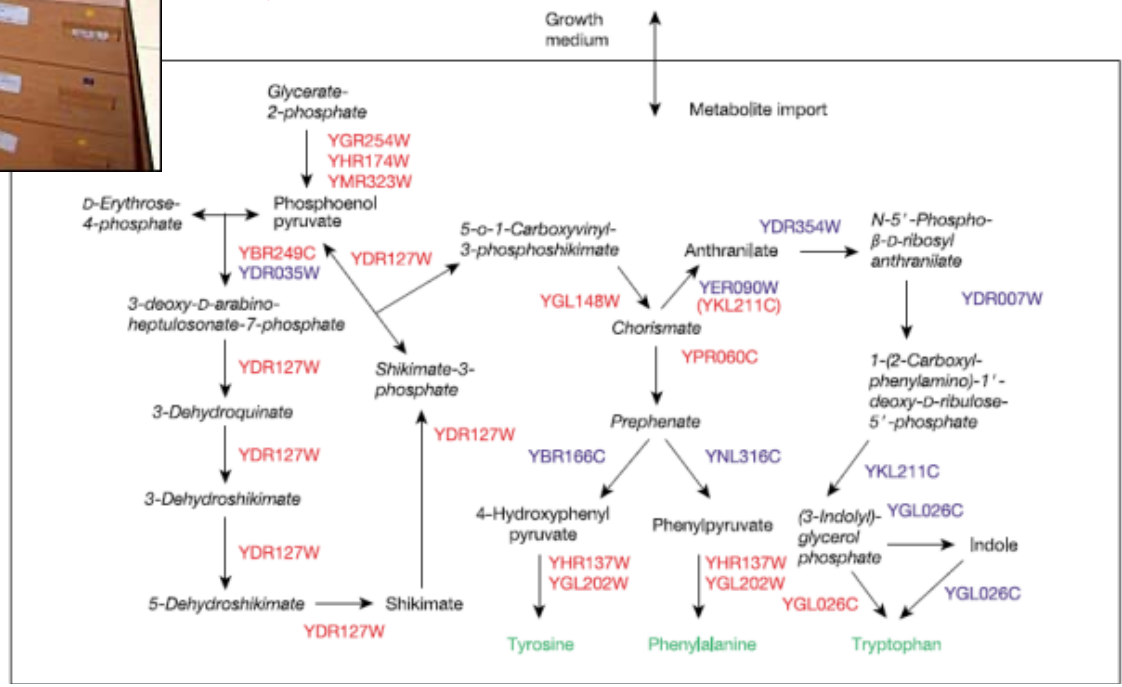
Voliro

[https://www.wysszurich.uzh.ch/projects/wyss-zurich-projects/voliro]

Robot Scientists

[King et al, Nature ’04, Science ‘09]





Autonomous Driving



CMU's Boss



Stanford's Stanley

- 1994, >1000km 3-lane highway w heavy traffic
- DARPA Grand Challenges:
  - 2005: drive 150 mile in the Mojave desert
  - 2007: drive 60 mile in traffic in urban environment
- Tesla / Waymo / … self-driving car projects

https://www.cmu.edu/news/image-archive/Boss.jpg

Games



IBM’s Deep Blue wins 6 game match against Garry Kasparov (’97)

https://media.wired.com/photos/5932a09d2a990b06268abee5/master/pass/deep-blue-kasparov.jpg

Games





- Go:
  - 2008: MoGo beats Pro (8P) in 9x9 game
  - 2016: AlphaGo wins 5th final game against Lee Sedol
  - 2017: Alpha Zero learns to play go (chess, shogi) from scratch
- Poker:
  - 2017: DeepStack and Libratus outperform pro Texas No limit Hold'em
  - 2019: Pluribus outperform pros at 6 player no-limit Hold'em

Computer games / eSports





[TheVerge 6.8.'18]

- 2018: DeepMind's AlphaStar defeats top pro player
- 2019: OpenAI Five Bot defeats pro human experts in Dota 2

Why should I take the course "Probabilistic Artificial Intelligence" at ETH Zurich?

Here are several strong reasons you might want to take **Probabilistic Artificial Intelligence (PAI)** at ETH Zürich, along with what to expect and what you'll gain. If you tell me your background (e.g. in probability, ML, maths), I can say more precisely how it would suit you.

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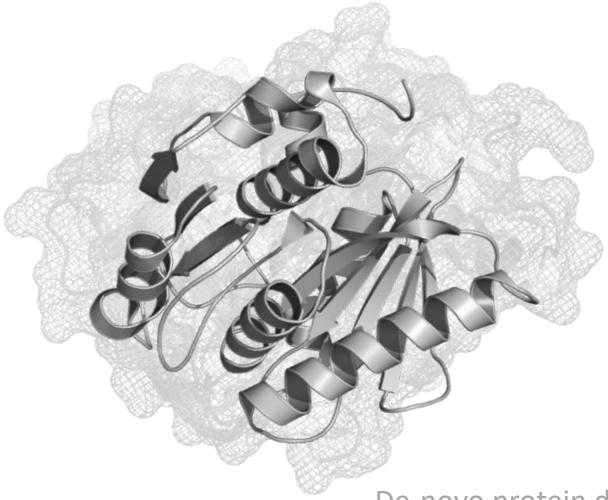
Why taking it is beneficial

1. **Handling uncertainty & real-world complexity**
   In many real-world AI / ML problems, data is noisy, incomplete, or uncertain. Probabilistic models provide tools to not just make predictions but also to *quantify how uncertain* those predictions are. That's critical in safety-sensitive domains (robotics, autonomous systems, medical diagnostics, etc.). The course explicitly focuses on managing uncertainty. arXiv +2
2. **Broader, deeper methodological foundations**
   You'll learn more than "just train a neural network". The course exposes you to statistical inference, planning under uncertainty, optimization, Bayesian methods, approximate inference, etc. These are powerful tools that generalize well across problem domains.
3. **Hands-on experience**
   With code projects, you'll both implement and experiment with these methods, not just learn them abstractly. That helps solidify understanding, which is especially important when dealing with probabilistic / approximate algorithms that are tricky in practice.
4. **Better decision-making & modelling skills**
   Probabilistic thinking sharpens modelling discipline — thinking about what's known vs unknown; what assumptions are required; balance between model complexity and data; trade-offs in inference (accuracy vs computational cost). These skills are valuable in research and industry alike.
5. **Good preparation for advanced AI / ML fields**
   If you want to do research (or advanced work) in areas like Bayesian deep learning, reinforcement learning, uncertainty quantification, robotics, autonomous agents, or any domain with sequential decision making, this course gives foundational tools.
6. **ETH Zürich's reputation & quality**
   ETH is among the top in AI / ML globally. Taking a rigorous course like this not only gives you good knowledge, but it's also recognized. And the professors / TAs are likely at the frontiers, exposing you to recent advances.

Generative AI

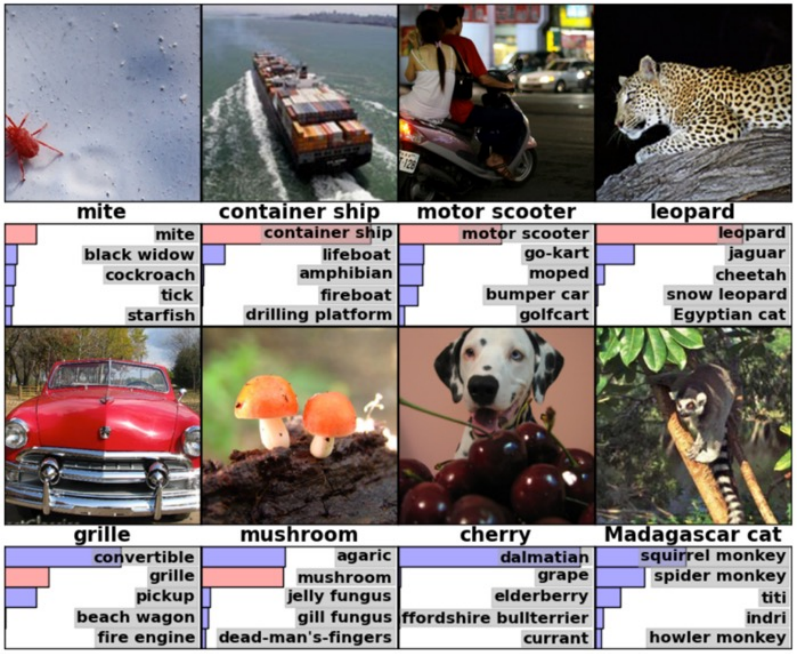


"Photorealistic closeup video of two pirate ships battling each other as they sail inside a cup of coffee" OpenAI Sora



De-novo protein design & structure prediction

Advances in Deep Learning

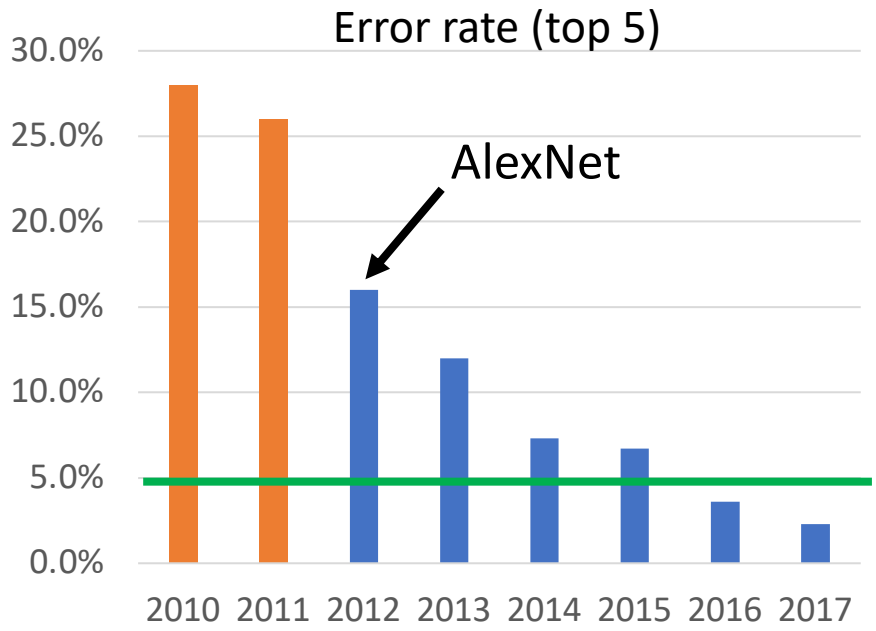


[Krizhevsky, Sutskever, Hinton. ImageNet Classification with Deep Convolutional Neural Networks, NeurIPS 2012]

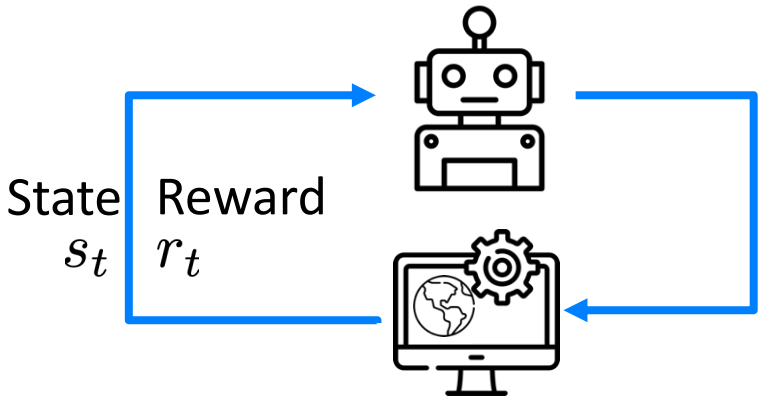


Source: ACM

2018 Turing Award
Bengio, Hinton & LeCun



Advances in Reinforcement Learning





Source: ACM

2024 Turing Award
Barto & Sutton



2013



2016



2023



2022

Challenges and Risks with Advanced AI

- What will happen to our jobs?
- What about misuse of AI?
- How to ensure AI systems remain “aligned”?
- Will machines surpass human intelligence?
- What will we do with superintelligent machines?
- What will they do with us?
- …



Topics covered

- **Probabilistic foundations of AI**

  ➔ learning and decision making under uncertainty

- Probabilistic machine learning (GPs, Bayesian deep learning, variational inference, diffusion models)
- Bandits & Bayesian optimization
- Planning under uncertainty (MDPs, POMDPs)
- (Deep) Reinforcement learning
- Applications (in class and in project)

Other ML & AI Courses @ ETH Zurich

- Deep Learning (Fall)
- Reliable and Trustworthy AI (Fall)
- Natural Language Processing (Fall)
- Guarantees for Machine Learning (Fall)

- Foundations of Reinforcement Learning (Spring)
- Large Language Models (Spring)
- Large-scale AI Engineering (Spring)
- Computational Intelligence Lab (Spring)
- …

Overview

- *Instructor*:
  Andreas Krause (krausea@ethz.ch)

- *Teaching assistants*: (pai26-info@inf.ethz.ch)
  Frederike Lübeck – *Head TA*

  Daniel Adorno, Mert Albaba, Yarden As, Anton Baumann, Jasmine Bayrooti, Xin Chen, Jin Cheng, Malte Franke, Jakub Frechowicz, Lino Hofstetter, Kornel Ipacs, Klemens Iten, Arghavan Kassraie, Jan Kogler, Bruce Lee, Chenhao Li, Frederieke Lohmann, Nicolas Menet, Laura Mismetti, Arad Mohammadi, Patrik Okanovic, Tomasz Puczel, Pierre Roth, Maximilian Seeliger, Daniel Simões Marta, Anuj Srivastava, Filippo Staffoni, Daniel Steinhauser, Tomasz Sternal, Scott Sussex, Balázs Szeker, Bartosz Szostakiewicz, Manuel Wendl, Patrik Wolf, Daniel Yang.

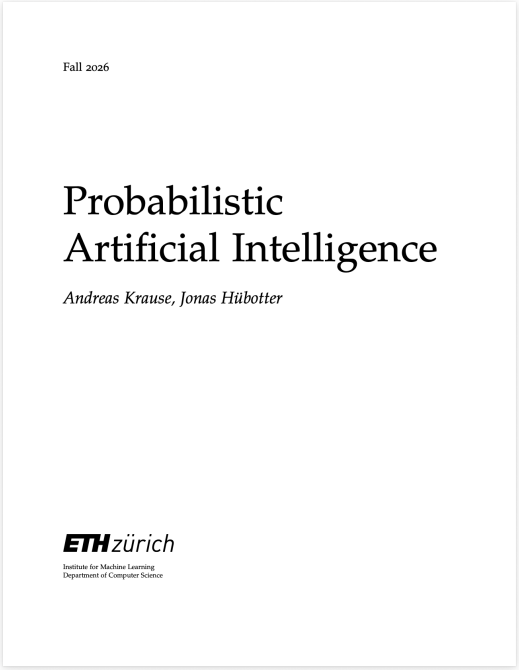
- *Administrative assistant*:
  Banu Büyüksahin (banu.bueyueksahin@inf.ethz.ch)

Course material

- Slides & information available on course webpage: https://las.inf.ethz.ch/teaching/pai-f26

- Lecture notes https://las.inf.ethz.ch/courses/pai-f26/script/main.pdf

  - Feedback appreciated via issue/merge requests: https://gitlab.inf.ethz.ch/OU-KRAUSE/pai-script



Further Reading

- Relevant books:
  - S. Russell, P. Norvig: Artificial Intelligence, A Modern Approach (4th edition)
  - C. Rasmussen, C.K.I. Williams: Gaussian Processes in Machine Learning http://www.gaussianprocess.org/gpml/
  - K. Murphy: Machine Learning: A Probabilistic Perspective
  - R. Sutton, A. Barto: Reinforcement Learning http://incompleteideas.net/book/RLbook2020.pdf
  - C. Szepesvári: Algorithms for Reinforcement Learning https://sites.ualberta.ca/~szepesva/papers/RLAlgsInMDPs.pdf
  - T. Lattimore, C. Szepesvári: Bandit Algorithms https://tor-lattimore.com/downloads/book/book.pdf
- Articles referenced on webpage

Background & Prequisites

- **Required**:
  - Solid basic knowledge in probability, linear algebra, calculus, algorithms and programming.
  - Introduction to Machine Learning (or similar)
    Material online: https://las.inf.ethz.ch/teaching/introml-s26

- We review necessary background, but will move quickly…

Coursework

- Grade based on written **session exam**
  (need to pass course project to be admitted to exam)

- **~ Six homeworks** (not graded)
  - Primarily using the Moodle environment
- **Exercise sessions:** Thursdays 16-18 (HG F7 + recorded)
  - Reviewing relevant material
  - Discussion of homework solutions
  - **Will start next week**

Lecture Format

- Physical attendance strongly encouraged
  - In principle possible to attend the class (lectures, exercises, projects) virtually (except exam)
- Opportunities for asking questions:
  - During class / breaks: in person, via EduApp chat (https://eduapp-app1.ethz.ch)
  - During virtual Q&A session on Mondays 17:15-18:00 (Zoom link on course webpage)
  - Asynchronously: Moodle forum

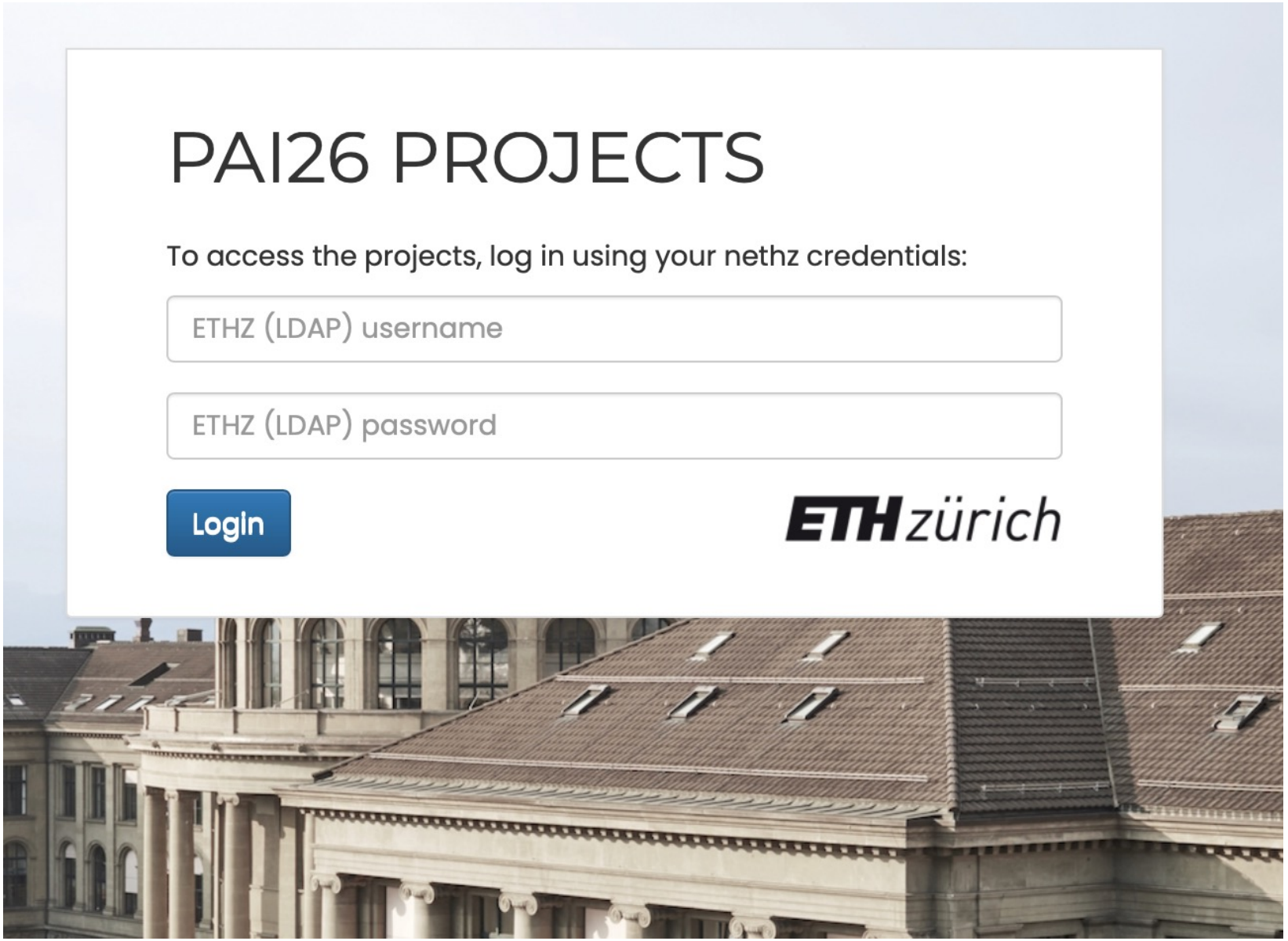
Course Project

- Hands on experience on Probabilistic machine learning, Bayesian optimization & Reinforcement learning
- Teams of (up to) 3 students
- Four smaller projects (+1 ungraded warmup)
- Details and grading (pass/fail) on https://las.inf.ethz.ch/wp-content/uploads/2026/09/pai26projectinfo.pdf
- 3 out of 4 projects must be passed for exam admission
- Best project award for each task
- Possible solutions discussed during Q&A session

Note on AI Use

- Manual implementation is encouraged to aid understanding
- AI tools may be used but must be declared
- Everyone must be able to fully understand and explain their solution in a video (and potential follow-up questions).

https://project.las.ethz.ch



FAQ

- Waitlist
- Auditing
- Doctoral students / “Testat”
- No live streaming but recording
- Remote exams

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Probabilistic Artificial Intelligence

Brief Recap of Probability

Prof. Andreas Krause
Learning and Adaptive Systems (las.ethz.ch)
Institute for Machine Learning

Review: Probability

- Formally: Probability Space $(\Omega, \mathcal{F}, P)$
  - Set of atomic events: Ω
  - Set of all non-atomic events: $\mathcal{F} \subseteq 2^{\Omega}$

    $\mathcal{F}$ is a σ-Algebra (closed under complements and countable unions)

$$\Omega \in \mathcal{F}$$

$$A \in \mathcal{F} \Rightarrow \Omega \setminus A \in \mathcal{F}$$

$$A_1, \ldots, A_n, \cdots \in \mathcal{F} \Rightarrow \bigcup_i A_i \in \mathcal{F}$$

  - Probability measure $P\colon \mathcal{F} \to [0, 1]$

    For $A \in \mathcal{F}, P(A)$ is the probability that event A happens

Probability Axioms

Normalization: $P(\Omega) = 1$

Non-negativity: $P(A) \geq 0 \text{ for all } A \in \mathcal{F}$

σ-additivity:

$$\forall A_1, \ldots A_n, \ldots \in \mathcal{F} \text{ disjoint}: P\left(\bigcup_{I=1}^{\infty} A_i\right) = \sum_{I=1}^{\infty} P(A_i)$$

Interpretation of probabilities

- Philosophical debate…
- Frequentist interpretation
  - $P(A)$ is relative frequency of A in repeated experiments
  - Can be difficult to assess with limited data
- Bayesian interpretation
  - $P(A)$ is "degree of belief" that A will occur
  - Where does this belief come from?
  - Many different flavors (subjective, objective, pragmatic, …)
- For now, assume probabilities are known

Random Variables

- Events are cumbersome to work with
- Let D be some set (e.g., the integers)
- A random variable X is a mapping $X: \Omega \to D$
- For some $x \in D$, we say

$$P(X = x) = P(\{\omega \in \Omega: X(\omega) = x\})$$

"probability that variable X assumes state x"

Specifying Probability Distributions through RVs

- Bernoulli distribution: "(biased) coin flips"

  $$D = \{H, T\}$$

  Specify $P(X = H) = p$. Then $P(X = T) = 1 - p$.

  Note: can identify atomic ev. ω with $\{X = H\}, \{X = T\}$

- Binomial distribution counts no. heads S in n flips

- Categorical distribution: "(biased) m-sided dice"

  $$D = \{1, \ldots, m\}$$

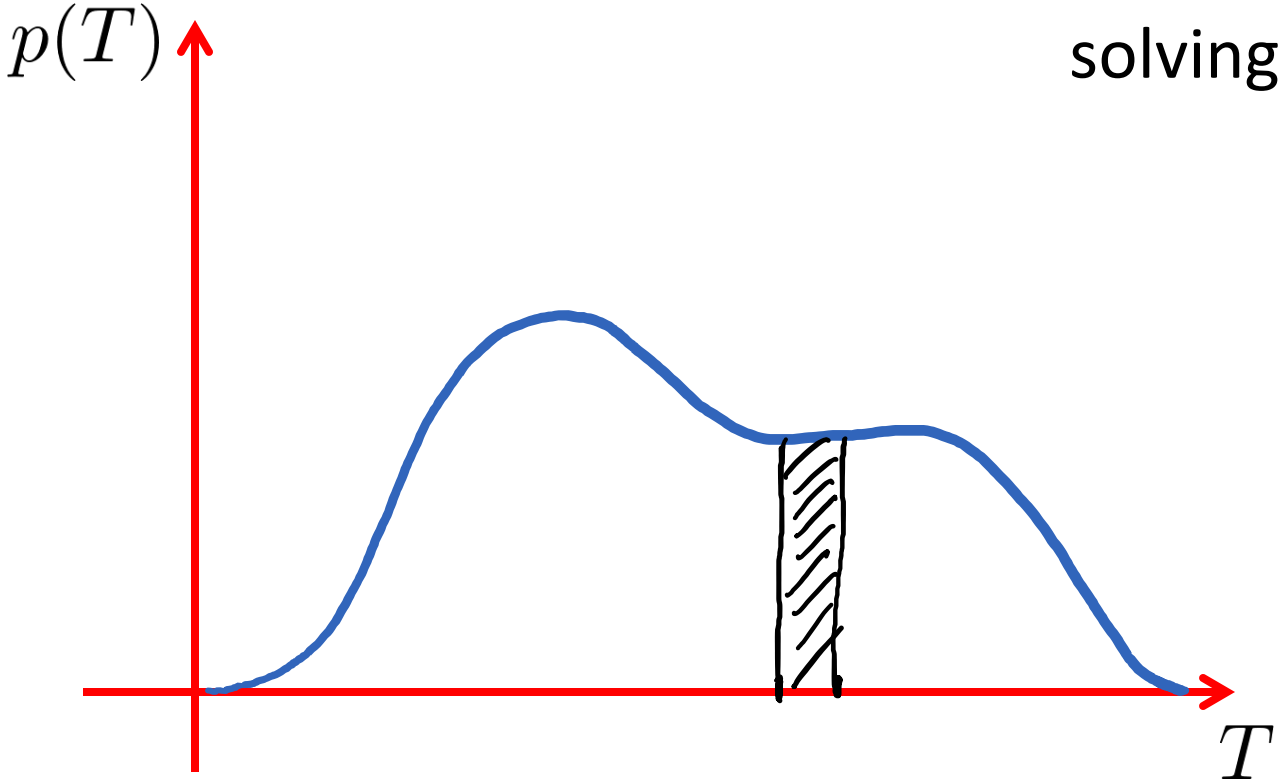
  Specify $P(X = i) = p_i$, s.t. $p_i \geq 0, \sum_i p_i = 1$

- Multinomial distribution counts the number of outcomes for each side for n throws

Continuous Distributions

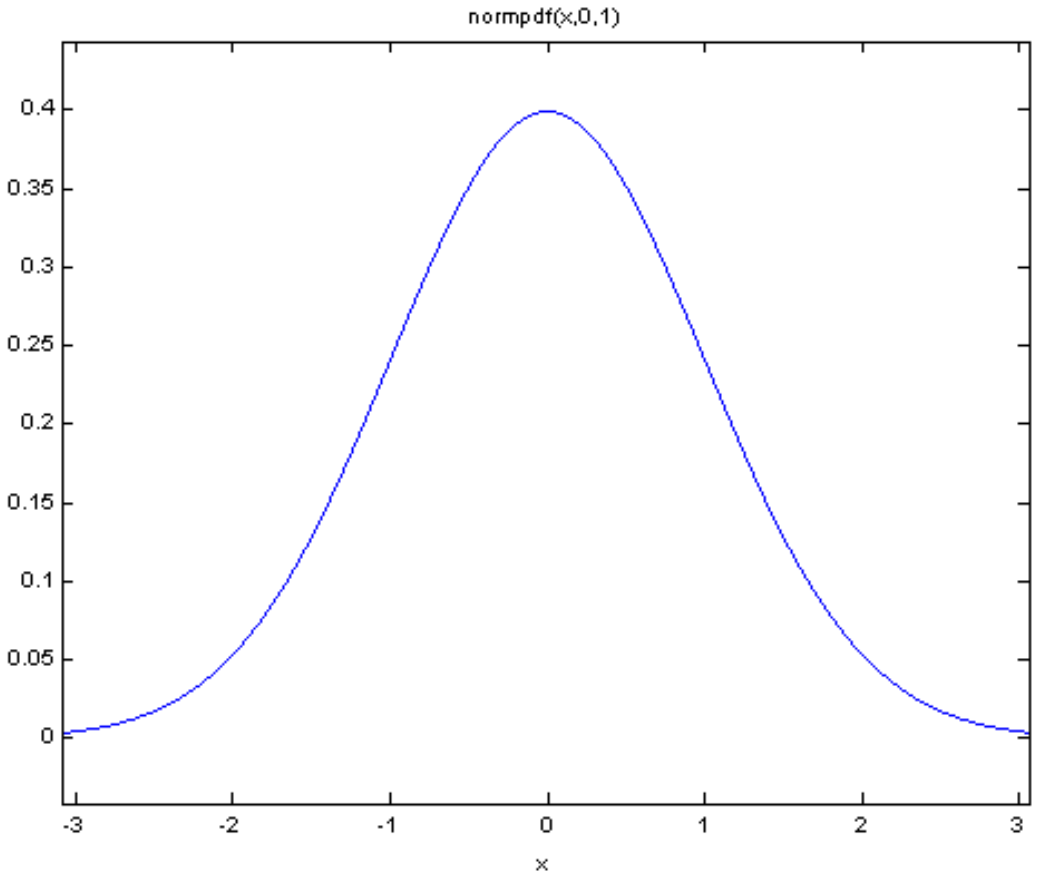
- Probability *density*

T = time taken for solving the homework



$$p(t) = \lim_{\delta \to 0} \frac{P([T \in [t, t+\delta)])}{\delta}$$

Example: Gaussian Distribution



- σ = Std. dev.
- μ = mean

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Expected values

- Given: Random variable X on domain D, and $f: D \to \mathbb{R}$
- Then expectation $\mathbb{E}[f(X)]$ is defined as:

$$\mathbb{E}[f(x)] = \begin{cases} \sum_{x \in D} f(x)\, p(x) & \text{if finite} \\ \int_D f(x)\, p(x)\, dx & \text{if continuous} \end{cases}$$

Eg. Mean: $\mu_X = \mathbb{E}[X] = \int x\, p(x)\, dx$

Var: $\sigma_X^2 = \mathbb{E}[(X-\mu)^2] = \int (x-\mu)^2 p(x)\, dx$

Joint Distributions

- Instead of random variable, have random vector

$$\mathbf{X} = [X_1(\omega), \ldots, X_n(\omega)]$$

- Can specify $P(X_1 = x_1, \ldots, X_n = x_n)$ directly (atomic events are assignments $x_1, \ldots, x_n$)
- Joint distribution describes relationship among all variables
- Example:

	toothache		no toothache	
	catch	no catch	catch	no catch
cavity	.108	.012	.072	.008
no cavity	.016	.064	.144	.576

P(toothache) = .108 + .012 + .016 + .064 = .2

Conditional Probability

- Formal definition:

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)} \text{ if } P(b) \neq 0$$

- Product rule $P(a \wedge b) = P(a \mid b)P(b)$
- For random vars: $P(A, B) = P(A \mid B)P(B)$
  (set of equations, one for each realization of A, B)

$$\forall a, b: \; P(A=a, B=b) = P(A=a \mid B=b) \cdot P(B=b)$$

- Product rule for multiple RVs:

$$P(X_{1:n}) = P(X_1, \ldots, X_n) = P(X_1)\, P(X_2 \mid X_1) \cdot P(X_3 \mid X_{1:2}) \cdots P(X_n \mid X_{1:n-1})$$

$$= P(X_n) \cdot P(X_{n-1} \mid X_n) \cdots P(X_1 \mid X_{2:n})$$

The Two Rules for Joint Distributions

Sum rule (Marginalization)

(A) $$P(X_{1:i-1}, X_{i+1:n}) = \sum_{x_i} P(X_{1:i-1}, x_i, X_{i+1:n})$$

$$p(x_{1:i-1}, x_{i+1:n}) = \int p(x_{1:n})\, dx_i$$

Product rule (chain rule)

$$P(X_1, \ldots, X_n) = P(X_1)P(X_2 \mid X_1) \ldots P(X_n \mid X_1, \ldots, X_{n-1})$$

(A) $\Rightarrow$ $$p(X_i) = \sum_{x_1} \cdots \sum_{x_{i-1}} \sum_{x_{i+1}} \cdots \sum_{x_n} p(x_{1:i-1}, X_i, x_{i+1:n})$$

Posterior Inference

- Suppose we know:
  - Prior probability $P(C)$

cavity	no cavity
.1	.9

  - Likelihood $P(T \mid C)$

	toothache	no toothache
cavity	.9	.1
no cavity	.01	.99

- How do we get $P(\text{cavity} \mid \text{toothache})$?

Bayes' Rule

Given:

- Prior P(X)
- Likelihood P(Y | X)



$$P(X \mid Y) \overset{\text{def}}{=} \frac{P(X,Y)}{P(Y)} \overset{\text{prod. rule}}{=} \frac{P(X) \cdot P(Y \mid X)}{\sum_x P(Y, X=x)} = \frac{P(X) \cdot P(Y \mid X)}{\sum_x P(Y \mid X=x) \cdot P(X=x)}$$

sum rule; prod. rule

Bayes' rule computes posterior

$$P(X \mid Y) = \frac{P(X)P(Y \mid X)}{\sum_{X=x} P(X=x)P(Y \mid X=x)}$$

Independent RVs

- Random variables $X_1 \ldots X_n$ are called independent if

$$P(X_1 = x_1, \ldots, X_n = x_n) = P(x_1)P(x_2)\cdots P(x_n)$$

$P(X,Y) = P(X) \cdot P(Y|X)$

$P(X,Y) = P(X) \cdot P(Y)$

$\Rightarrow P(Y|X) = P(Y)$ as long as $P(X) > 0$

- Independence is a very strong requirement.
  Is there something weaker?

Conditional Independence (example)

- If I know there's a *cavity*, knowing *toothache* won't help predict whether the probe *catches*

$$P(\text{catch} \mid \text{cavity}, \text{toothache}) = P(\text{catch} \mid \text{cavity})$$

- for all values of *catch, cavity* and *toothache*

Key Concept: Conditional Independence

- Rand. vars. X and Y conditionally independent given Z iff for all x, y, z :

$$P(X = x, Y = y \mid Z = z) = P(X = x \mid Z = z)P(Y = y \mid Z = z)$$

- If $P(Y = y \mid Z = z) > 0$, that is equivalent to

$$P(X = x \mid Z = z, Y = y) = P(X = x \mid Z = z)$$

Similarly for sets of random variables $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$

We write:

$$\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}$$

Problems with High-dim. Distributions

- Suppose we have n binary variables
- How many parameters do we need to specify $P(X_1 = x_1, \ldots, X_n = x_n)$?

X_1	X_2	…	X_{n-1}	X_n	P(X)
0	0	…	0	0	.01
0	0	…	1	0	.001
0	0	…	1	1	.213
…	…	…	…	…	
1	1	…	1	1	.0003

2^n-1 parameters!



More Problems: Computing Marginals

- Suppose we have joint distribution $P(X_1, \ldots, X_n)$
- Then (acc. to sum rule)

$$P(X_i = x_i) = \sum_{x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n} P(x_1, \ldots, x_n)$$

- If all X_i are binary: How many terms does this sum have?

2^{n-1} !!

More Problems: Conditional Queries

- Suppose we have joint distribution $P(X_1, \ldots, X_n)$
- Compute distribution of some variables given values for others:

$$P(X_1 = \cdot \mid X_7 = x_7) = \frac{P(X_1 = \cdot, X_7 = x_7)}{P(X_7 = x_7)} = \frac{1}{Z} P(X_1 = \cdot, X_7 = x_7)$$

$$Z = \sum_{x'} P(X_1 = x', X_7 = x_7)$$

$$P(X_1, X_2 = x_7) = \sum_{x_2} \sum_{x_3, x_6} \sum_{x_8 \ldots x_n} P(X_1, x_2 \ldots x_n)$$

Challenges with high-dimensional distributions

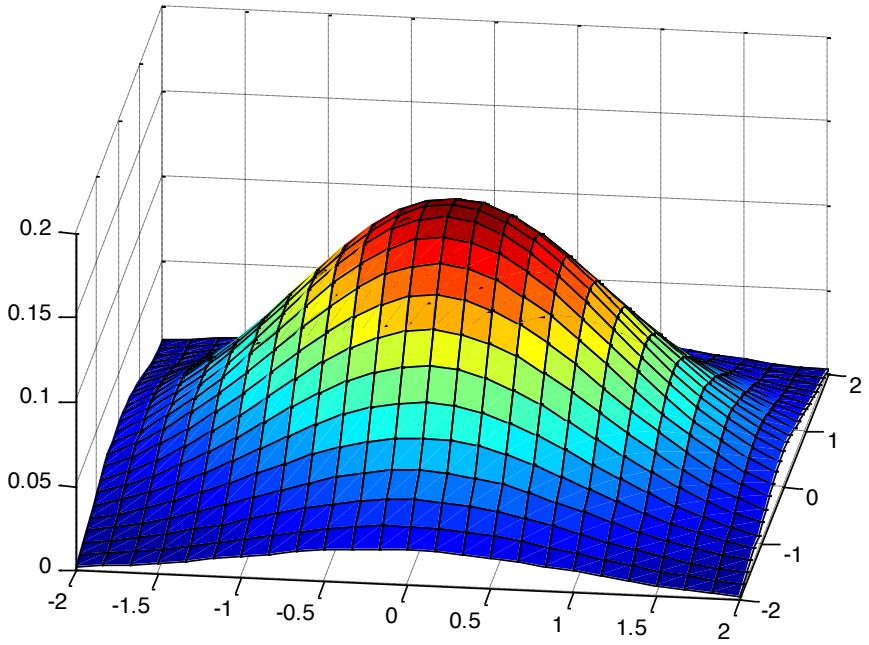
- Representation (parametrization)
- Learning (estimation)
- Inference (prediction)

- In the following, we will start with a particular family of distributions where these problems are tractable
  - Multivariate Gaussians

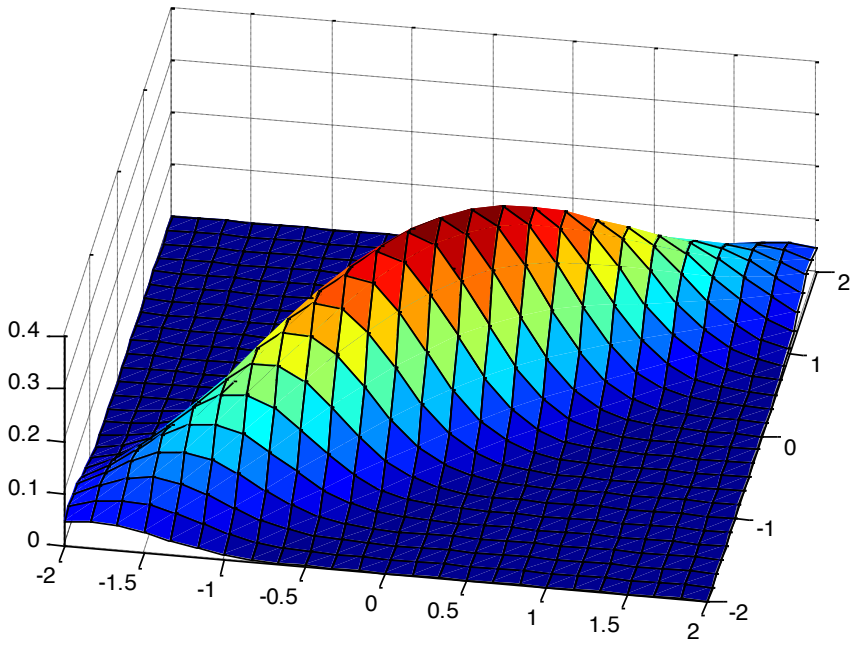
- We will discuss how to do Bayesian learning and inference with Gaussians (and Gaussian processes)
- Then generalize to more complex models, approximate inference etc. (Bayesian neural nets etc.)

Example: Multivariate Gaussian

$$p(x) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right) \qquad \Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix} \qquad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$$

Multivariate Gaussian distribution

$$N(x;\mu,\Sigma) = \frac{1}{(2\pi)^{n/2}\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \vdots & & & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix}$$

$\mu = [\mu_1 \dots \mu_n]^T$ mean vector

$\mu_i = \mathbb{E}[X_i]$

$\sigma_i^2 = \mathrm{Var}[X_i] = \mathbb{E}[(X_i - \mu_i)^2]$

$\sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]$

Covariance matrix Σ is symmetric and positive semidefinite

Thus joint distribution over n variables requires only $O(n^2)$ parameters!

Fact: Gaussians are independent iff they are uncorrelated

Marginal inference in Gaussian Distributions

- Suppose we have a Gaussian random vector

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$

- Hereby $V = \{1,...,d\}$ is an index set.
- Suppose we consider a subset of the variables

$$A = \{i_1, \ldots, i_k\} \qquad A \subset V$$

- The marginal distribution of variables indexed by *A* is:

$$\mathbf{X}_A = [X_{i_1}, \ldots, X_{i_k}] \sim \mathcal{N}(\mu_A, \Sigma_{AA})$$

$\mu_A = [\mu_{i_1} \ldots \mu_{i_k}]^T$ $\Sigma_{AA} = \begin{pmatrix} \sigma^2_{i_1} & \sigma_{i_1 i_2} & \cdots & \sigma_{i_1 i_s} \\ \vdots & & \ddots & \\ \sigma_{i_1 i_k} & \cdots & & \sigma^2_{i_k} \end{pmatrix}$

Conditional distributions

- Suppose we have a Gaussian random vector

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$

- Further, suppose we take two disjoint subsets of V

$$A = \{i_1, \ldots, i_k\} \qquad B = \{j_1, \ldots, j_m\}$$

- The conditional distribution

$$p(\mathbf{X}_A \mid \mathbf{X}_B = \mathbf{x}_B) = \mathcal{N}(\mu_{A|B}, \Sigma_{A|B})$$

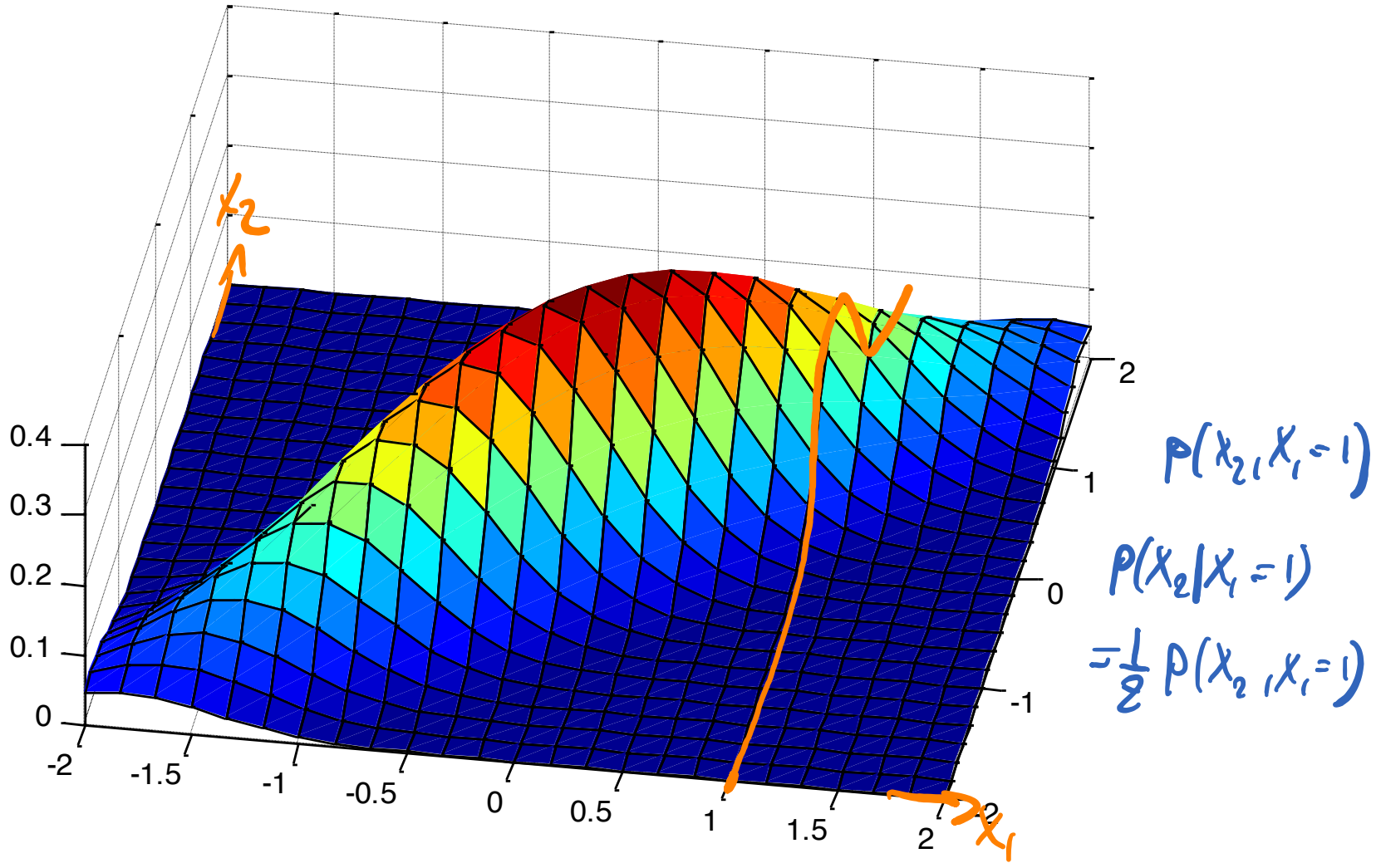
is Gaussian, where

$$\Sigma_{AB} = \begin{pmatrix} \sigma_{i_1 j_1} & \cdots & \sigma_{i_1 j_m} \\ \vdots & & \vdots \\ \sigma_{i_k j_1} & \cdots & \sigma_{i_k j_m} \end{pmatrix}$$

$$\mu_{A|B} = \mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(\mathbf{x}_B - \mu_B)$$

$$\Sigma_{A|B} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$$

Illustration



Multiples of Gaussians are Gaussian

- Suppose we have a Gaussian random vector

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$

- Take a matrix $M \in \mathbb{R}^{m \times d}$
- Then the random vector $\mathbf{Y} = \mathbf{M}\mathbf{X}$ is Gaussian:

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{M}\mu_V, \mathbf{M}\Sigma_{VV}\mathbf{M}^T)$$

Sums of Gaussians are Gaussian

- Suppose we have two independent Gaussian random vectors

$$\mathbf{X} = \mathbf{X}_V = [X_1, \ldots, X_d] \sim \mathcal{N}(\mu_V, \Sigma_{VV})$$
$$\mathbf{X}' = \mathbf{X}'_V = [X'_1, \ldots, X'_d] \sim \mathcal{N}(\mu'_V, \Sigma'_{VV})$$

- Then the random vector $\mathbf{Y} = \mathbf{X} + \mathbf{X}'$ is Gaussian:

$$\mathbf{Y} \sim \mathcal{N}(\mu_V + \mu'_V, \Sigma_{VV} + \Sigma'_{VV})$$

Sps. c is a constant, X Gaussian, $X \sim N(\mu, \sigma^2)$

$X + c \sim N(\mu + c, \sigma^2)$ (i.e. interpreting $c \sim N(c, 0)$

Conditional Linear Gaussians

- If X, Y are jointly Gaussian, then $p(X = x \mid Y = y)$ is Gaussian, with mean linearly dependent on *y*

$$p(X \mid Y=y) = \mathcal{N}(\mu_{X|Y=y}, \sigma^2_{X|Y=y}), \quad \mu_{X|Y=y} = \mu_X + \sigma_{XY}\sigma_Y^{-2}(y - \mu_Y)$$

$$\sigma^2_{X|Y=y} = \sigma_X^2 - \sigma_{XY}^2\sigma_Y^{-2}$$

- Thus random variable X can be viewed as a linear function of Y with independent Gaussian noise added

$$\Rightarrow Y = a \cdot X + b + \varepsilon, \quad a = \sigma_{XY}\sigma_Y^{-2}, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2_{X|Y=y})$$

$$b = \mu_X + \sigma_{XY}\sigma_Y^{-2}\mu_Y$$

- The converse also holds.

Conditional Linear Gaussians

- If X, Y are jointly Gaussian, then $p(X = x \mid Y = y)$ is Gaussian, with mean linearly dependent on y

$$p(X = x \mid Y = y) = N\left(x; \mu_{X|Y=y}, \sigma^2_{X|Y}\right)$$
$$\mu_{X|Y=y} = \mu_X + \sigma_{XY}\sigma_Y^{-2}(y - \mu_Y)$$
$$\sigma^2_{X|Y} = \sigma^2_X - \sigma^2_{XY}\sigma_Y^{-2}$$

- Thus random variable X can be viewed as a linear function of Y with independent Gaussian noise added

$$X = aY + b + \epsilon, \text{ where } \epsilon \sim N(0, \sigma^2_{X|Y})$$
$$\text{and } a = \sigma_{XY}\sigma_Y^{-2}, b = \mu_X - \sigma_{XY}\sigma_Y^{-2}\mu_y$$

- The converse also holds.

Outlook

- Multivariate Gaussians have important properties:
  - Compact representation of high-dimensional joint distributions
  - Closed form inference

- In the following, will discuss how we can make use of these properties for Bayesian learning

Reading

- Chapter 1 in the lecture notes